

## TRENTON-MERCER, NJ, USA

WMO: 724095

Lat: 40.277N

Lon: 74.816W

Elev: 190

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14792

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.6       | 15.7 | -6.0                          | 4.0 | 14.2 | -1.2 | 5.2 | 18.5 | 23.7                  | 35.0 | 20.8 | 34.0 | 8.2                   | 320  | 0.464 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.2                   | 92.3            | 74.4 | 89.8 | 73.2 | 87.3 | 72.3 | 77.3                | 87.4 | 75.9 | 84.8 | 74.8 | 82.8 | 8.5                  | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.4                            | 129.6 | 81.4 | 73.1 | 124.0 | 79.8 | 72.3 | 120.2 | 79.1 | 40.7          | 87.5 | 39.4 | 85.0 | 38.3 | 83.0 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 20.0              | 17.8 | 16.0 | DB                         | 6.7 | 96.2               | 4.8 | 3.8                                                | 3.2 | 98.9       | 0.5  | 101.2      | -2.2 | 103.3      | -5.7 | 106.1 |
|                   |      |      | WB                         | 4.7 | 79.9               | 4.4 | 1.4                                                | 1.5 | 80.9       | -1.1 | 81.7       | -3.5 | 82.5       | -6.7 | 83.6  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 54.7   | 32.0  | 34.6  | 42.1  | 52.9  | 62.7  | 71.3  | 76.5  | 74.8  | 67.9  | 56.3  | 46.2  | 37.4 |
|                                                                               | DBStd       | 17.31  | 10.11 | 9.14  | 9.19  | 8.35  | 7.56  | 6.42  | 5.12  | 5.10  | 6.54  | 8.00  | 8.12  | 9.10 |
|                                                                               | HDD50       | 1943   | 564   | 439   | 279   | 59    | 2     | 0     | 0     | 0     | 0     | 28    | 167   | 405  |
|                                                                               | HDD65       | 4873   | 1024  | 852   | 710   | 375   | 139   | 19    | 0     | 2     | 43    | 289   | 564   | 856  |
|                                                                               | CDD50       | 3645   | 4     | 6     | 34    | 146   | 395   | 640   | 821   | 770   | 538   | 224   | 53    | 14   |
|                                                                               | CDD65       | 1103   | 0     | 0     | 1     | 12    | 66    | 209   | 356   | 307   | 131   | 20    | 1     | 0    |
|                                                                               | CDH74       | 9647   | 0     | 0     | 10    | 140   | 658   | 1842  | 3363  | 2574  | 937   | 122   | 1     | 0    |
|                                                                               | CDH80       | 3384   | 0     | 0     | 0     | 33    | 193   | 624   | 1352  | 904   | 259   | 19    | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.9    | 7.6   | 7.8   | 8.5   | 8.2   | 6.8   | 6.3   | 5.8   | 5.5   | 6.1   | 6.5   | 6.8   | 7.2  |
| Precipitation                                                                 | PrecAvg     | 42.3   | 3.1   | 3.0   | 3.6   | 3.1   | 3.4   | 3.4   | 4.9   | 3.9   | 4.2   | 2.8   | 3.1   | 3.7  |
|                                                                               | PrecMax     | 59.4   | 9.0   | 5.2   | 7.3   | 6.6   | 6.2   | 6.3   | 13.4  | 9.6   | 9.1   | 5.4   | 7.8   | 7.0  |
|                                                                               | PrecMin     | 30.4   | 0.4   | 1.0   | 0.9   | 0.8   | 0.2   | 0.6   | 1.3   | 0.5   | 0.8   | 0.0   | 0.3   | 0.6  |
|                                                                               | PrecStd     | 7.6    | 1.9   | 1.2   | 1.5   | 1.4   | 1.7   | 1.8   | 3.2   | 2.4   | 2.3   | 1.4   | 2.0   | 1.9  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 63.6  | 66.6  | 74.5  | 85.3  | 89.3  | 93.1  | 97.4  | 95.0  | 91.6  | 82.2  | 71.8  | 67.1 |
|                                                                               |             | MCWB   | 58.0  | 56.7  | 59.9  | 66.3  | 71.2  | 74.8  | 74.6  | 75.3  | 73.4  | 68.5  | 62.4  | 60.6 |
|                                                                               | 2%          | DB     | 57.4  | 59.6  | 67.1  | 78.8  | 85.4  | 89.9  | 93.1  | 91.0  | 86.3  | 77.3  | 67.1  | 61.3 |
|                                                                               |             | MCWB   | 52.7  | 52.2  | 55.8  | 61.9  | 69.2  | 73.2  | 74.5  | 74.3  | 71.9  | 67.4  | 59.4  | 56.8 |
|                                                                               | 5%          | DB     | 51.0  | 53.4  | 62.1  | 72.9  | 81.4  | 86.6  | 90.2  | 88.0  | 82.3  | 73.1  | 63.3  | 55.8 |
|                                                                               |             | MCWB   | 46.1  | 45.8  | 51.9  | 59.2  | 67.3  | 72.0  | 73.7  | 72.8  | 69.8  | 64.7  | 56.3  | 51.6 |
|                                                                               | 10%         | DB     | 46.1  | 47.8  | 56.9  | 68.0  | 77.2  | 83.5  | 87.5  | 84.9  | 79.4  | 69.7  | 59.5  | 51.2 |
|                                                                               |             | MCWB   | 41.5  | 41.4  | 48.5  | 56.0  | 65.1  | 69.9  | 72.7  | 71.3  | 68.3  | 62.1  | 53.7  | 46.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 59.7  | 60.1  | 62.1  | 68.8  | 73.6  | 77.2  | 79.4  | 79.2  | 76.6  | 72.6  | 65.2  | 62.2 |
|                                                                               |             | MCDB   | 62.3  | 65.2  | 71.6  | 79.0  | 84.8  | 88.1  | 91.8  | 89.8  | 85.5  | 77.7  | 68.7  | 65.4 |
|                                                                               | 2%          | WB     | 53.6  | 53.5  | 57.9  | 65.0  | 71.3  | 75.5  | 77.5  | 77.1  | 74.7  | 69.4  | 61.8  | 57.3 |
|                                                                               |             | MCDB   | 56.3  | 58.6  | 65.0  | 74.7  | 81.9  | 85.2  | 88.5  | 86.1  | 81.3  | 75.2  | 65.8  | 60.3 |
|                                                                               | 5%          | WB     | 46.7  | 46.2  | 53.4  | 61.8  | 69.3  | 74.3  | 76.1  | 75.6  | 73.1  | 66.7  | 58.2  | 52.6 |
|                                                                               |             | MCDB   | 50.0  | 51.3  | 59.7  | 70.2  | 78.6  | 83.2  | 85.6  | 83.3  | 78.8  | 71.9  | 62.0  | 55.6 |
|                                                                               | 10%         | WB     | 42.1  | 42.2  | 49.0  | 58.4  | 67.1  | 72.5  | 74.8  | 74.3  | 71.3  | 63.5  | 54.9  | 47.0 |
|                                                                               |             | MCDB   | 45.8  | 47.3  | 56.6  | 66.0  | 74.5  | 80.5  | 83.3  | 81.2  | 76.8  | 67.8  | 58.4  | 50.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.3  | 15.8  | 17.3  | 19.5  | 19.1  | 18.2  | 18.2  | 17.4  | 18.1  | 17.7  | 16.9  | 14.0 |
|                                                                               |             | MCDBR  | 21.1  | 23.5  | 25.2  | 26.6  | 24.4  | 21.3  | 21.2  | 20.2  | 21.3  | 21.1  | 21.1  | 19.3 |
|                                                                               |             | MCWBR  | 17.9  | 17.9  | 16.4  | 14.3  | 11.7  | 9.2   | 7.7   | 7.6   | 9.7   | 12.0  | 14.8  | 17.2 |
|                                                                               | 5% WB       | MCDBR  | 20.0  | 21.9  | 22.8  | 22.8  | 21.3  | 18.9  | 17.9  | 16.7  | 16.9  | 17.3  | 17.8  | 18.2 |
|                                                                               |             | MCWBR  | 18.2  | 18.5  | 17.3  | 14.5  | 11.1  | 9.4   | 7.7   | 7.5   | 9.4   | 12.0  | 14.8  | 17.9 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.308  | 0.320 | 0.358 | 0.387 | 0.433 | 0.496 | 0.492 | 0.485 | 0.400 | 0.358 | 0.327 | 0.307 |      |
|                                                                               | taud        | 2.464  | 2.421 | 2.381 | 2.330 | 2.227 | 2.093 | 2.119 | 2.125 | 2.338 | 2.462 | 2.511 | 2.521 |      |
|                                                                               | Ebn at Noon | 272    | 284   | 284   | 281   | 269   | 252   | 251   | 249   | 266   | 267   | 263   | 264   |      |
|                                                                               | Edh at Noon | 25     | 30    | 35    | 39    | 44    | 51    | 49    | 47    | 36    | 29    | 24    | 22    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 602    | 891   | 1177  | 1524  | 1689  | 1852  | 1895  | 1630  | 1362  | 963   | 682   | 519   |      |
|                                                                               | RadStd      | 53     | 95    | 95    | 140   | 193   | 158   | 91    | 106   | 149   | 100   | 61    | 52    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (68 neighbors) | +0.74 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +302  | +196  |
|      |                         |       |         |        |         |       |       |             |       | +214  | +122  |

Nomenclature: See separate page